Database Part – 1

Folder Structure:

```
L9
  .venv
  a1
  lab9
    __pycache__
    __init__.py
    asgi.py
    settings.py
    urls.py
    wsgi.py
  q1
    __pycache__
    migrations
    templates / q1
      index.html
    __init__.py
    admin.py
    apps.py
    forms.py
    models.py
    tests.py
    urls.py
    views.py
  q2
  db.sqlite3
  manage.py
```

Settings.py:

```
INSTALLED_APPS = [
'django.contrib.admin',
'django.contrib.auth',
'django.contrib.contenttypes',
'django.contrib.sessions',
'django.contrib.messages',
'django.contrib.staticfiles',
'q1',
'q2',
'a1',
```

```
]

DATABASES = {
'default': {
'ENGINE': 'django.db.backends.sqlite3',
'NAME': BASE_DIR / 'db.sqlite3',
}
}
```

Urls.py:

```
from django.contrib import admin
from django.urls import include, path

urlpatterns = [
path('admin/', admin.site.urls),
path('q1/', include('q1.urls')),
path('q2/', include('q2.urls')),
path('a1/', include('a1.urls')),
]
```

Q1:

Admin.py:

```
from django.contrib import admin

from .models import Category, Page

admin.site.register(Category)
admin.site.register(Page)
```

Models.py:

```
from django.db import models

class Category(models.Model):
name = models.CharField(max_length=100, unique=True)
visits = models.PositiveIntegerField(default=0)
likes = models.PositiveIntegerField(default=0)

def __str__(self):
return self.name



class Page(models.Model):
category = models.ForeignKey(Category, on_delete=models.CASCADE, related_name='pages')
```

```
title = models.CharField(max_length=150)
url = models.URLField(max_length=300)
views = models.PositiveIntegerField(default=0)

def __str__(self):
return self.title
```

Forms.py:

```
from django import forms

from .models import Category, Page


class CategoryForm(forms.ModelForm):
class Meta:
model = Category
fields = ['name', 'visits', 'likes']


class PageForm(forms.ModelForm):
class Meta:
model = Page
fields = ['category', 'title', 'url', 'views']
```

Urls.py:

```
from django.urls import path

from . import views

app_name = 'q1'

urlpatterns = [
path('', views.index, name='index'),
]
```

Views.py:

```
from django.shortcuts import redirect, render

from .forms import CategoryForm, PageForm
from .models import Category


def index(request):
category_form = CategoryForm(prefix='category')
```

```
page_form = PageForm(prefix='page')

if request.method == 'POST':
if 'add_category' in request.POST:
category_form = CategoryForm(request.POST, prefix='category')
if category_form.is_valid():
category_form.save()
return redirect('q1:index')
elif 'add_page' in request.POST:
page_form = PageForm(request.POST, prefix='page')
if page_form.is_valid():
page_form.save()
return redirect('q1:index')

categories = Category.objects.prefetch_related('pages').all().order_by('name')
context = {
'category_form': category_form,
'page_form': page_form,
'categories': categories,
}
return render(request, 'q1/index.html', context)
```

Index.html:

```
<!DOCTYPE html>
<html>
<head>
<title>Q1 - Website Directory</title>
</head>
<body>
<h1>Q1: Website Directory</h1>

<h2>Add Category</h2>
<form method="post">
{% csrf_token %}
{{ category_form.as_p }}
<button type="submit" name="add_category">Save Category</button>
</form>

<h2>Add Page</h2>
<form method="post">
{% csrf_token %}
{{ page_form.as_p }}
<button type="submit" name="add_page">Save Page</button>
</form>
```

```
<h2>Categories and Pages</h2>
{% for category in categories %}
<h3>{{ category.name }} (Visits: {{ category.visits }}, Likes: {{ category.likes }})</h3>
<ul>
{% for page in category.pages.all %}
<li>
{{ page.title }} - <a href="{{ page.url }}">{{ page.url }}</a> (Views: {{ page.views }})
</li>
{% empty %}
<li>No pages in this category.</li>
{% endfor %}
</ul>
{% empty %}
<p>No categories added yet.</p>
{% endfor %}
</body>
</html>
```

Screenshots:

Django administration

Home › Q1 › Categorys

Start typing to filter...

A1

Institutes + Add

AUTHENTICATION AND AUTHORIZATION

Groups + Add

Users + Add

Q1

Categorys + Add

Pages + Add

Select category to change

Action: --------- Go 0 of 1 selected

	CATEGORY
☐	Movies

1 category

Django administration

Home › Q1 › Pages

Start typing to filter...

A1

Institutes + Add

AUTHENTICATION AND AUTHORIZATION

Groups + Add

Users + Add

Q1

Categorys + Add

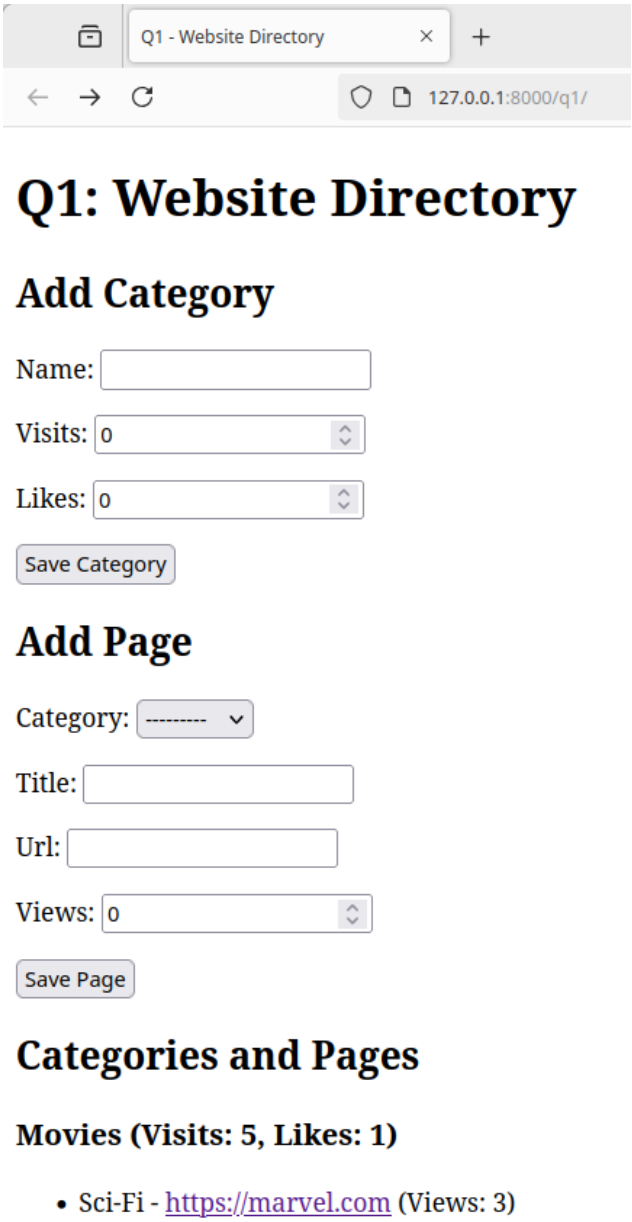
Pages + Add

Select page to change

Action: --------- Go 0 of 1 selected

	PAGE
☐	Sci-Fi

1 page



Q2:

Admin.py:

```
from django.contrib import admin

from .models import Lives, Works

admin.site.register(Works)
admin.site.register(Lives)
```

Models.py:

```
from django.db import models
```

```
class Lives(models.Model):
person_name = models.CharField(max_length=100, primary_key=True)
street = models.CharField(max_length=150)
city = models.CharField(max_length=100)

class Meta:
db_table = 'LIVES'

def __str__(self):
return f"{self.person_name} - {self.city}"


class Works(models.Model):
person_name = models.CharField(max_length=100)
company_name = models.CharField(max_length=100)
salary = models.DecimalField(max_digits=12, decimal_places=2)

class Meta:
db_table = 'WORKS'

def __str__(self):
return f"{self.person_name} @ {self.company_name}"
```

Forms.py:

```
from django import forms

from .models import Lives, Works

class LivesForm(forms.ModelForm):
class Meta:
model = Lives
fields = ['person_name', 'street', 'city']

class WorksForm(forms.ModelForm):
class Meta:
model = Works
fields = ['person_name', 'company_name', 'salary']

class CompanySearchForm(forms.Form):
company_name = forms.CharField(max_length=100)
```

Urls.py:

```
from django.urls import path
```

```
from . import views

app_name = 'q2'

urlpatterns = [
path('', views.index, name='index'),
]
```

Views.py:

```
from django.db.models import OuterRef, Subquery
from django.shortcuts import redirect, render

from .forms import CompanySearchForm, LivesForm, WorksForm
from .models import Lives, Works



def index(request):
lives_form = LivesForm(prefix='lives')
works_form = WorksForm(prefix='works')
search_form = CompanySearchForm(prefix='search')
results = None

if request.method == 'POST':
if 'add_lives' in request.POST:
lives_form = LivesForm(request.POST, prefix='lives')
if lives_form.is_valid():
lives_form.save()
return redirect('q2:index')
elif 'add_work' in request.POST:
works_form = WorksForm(request.POST, prefix='works')
if works_form.is_valid():
works_form.save()
return redirect('q2:index')
elif 'search_company' in request.POST:
search_form = CompanySearchForm(request.POST, prefix='search')
if search_form.is_valid():
company_name = search_form.cleaned_data['company_name']
lives_city_subquery = Lives.objects.filter(
person_name=OuterRef('person_name')
).values('city')[:1]
results = (
Works.objects.filter(company_name__iexact=company_name)
.annotate(city=Subquery(lives_city_subquery))
```

```
.values('person_name', 'city')
)

context = {
'lives_form': lives_form,
'works_form': works_form,
'search_form': search_form,
'results': results,
}
return render(request, 'q2/index.html', context)
```

Index.html:

```
<!DOCTYPE html>
<html>
<head>
<title>Q2 - WORKS and LIVES</title>
</head>
<body>
<h1>Q2: WORKS and LIVES</h1>

<h2>Insert into LIVES</h2>
<form method="post">
{% csrf_token %}
{{ lives_form.as_p }}
<button type="submit" name="add_lives">Insert</button>
</form>

<h2>Insert into WORKS</h2>
<form method="post">
{% csrf_token %}
{{ works_form.as_p }}
<button type="submit" name="add_work">Insert</button>
</form>

<h2>Find Employees by Company</h2>
<form method="post">
{% csrf_token %}
{{ search_form.as_p }}
<button type="submit" name="search_company">Search</button>
</form>

{% if results is not None %}
<h3>Results</h3>
<ul>
```

```
{% for row in results %}
<li>{{ row.person_name }} - {{ row.city|default:"City not found" }}</li>
{% empty %}
<li>No records found for this company.</li>
{% endfor %}
</ul>
{% endif %}
</body>
</html>
```

Screenshots:

Select works to change | Django

127.0.0.1:8000/admin/q2/works/

Django administration

Home › Q2 › Workss

Start typing to filter...

A1

Institutes + Add

AUTHENTICATION AND AUTHORIZATION

Groups + Add

Users + Add

Q1

Categorys + Add

Pages + Add

Q2

Livess + Add

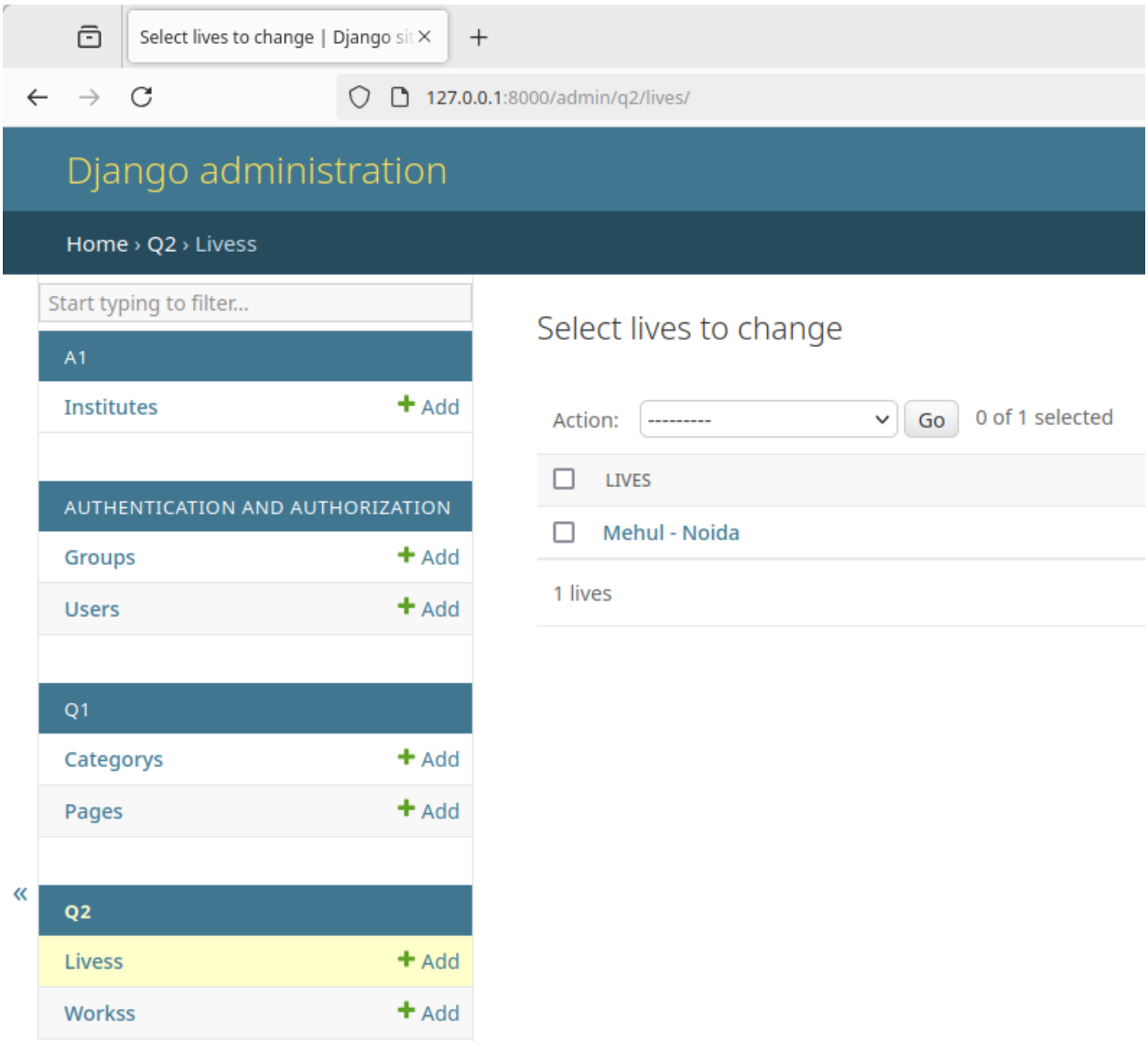
Workss + Add

Select works to change

Action: --------- Go 0 of 1 selected

	WORKS
☐	Mehul @ STM

1 works


Select lives to change | Django sit
127.0.0.1:8000/admin/q2/lives/
Django administration
Home › Q2 › Livess
Start typing to filter...
A1
Institutes
Add
AUTHENTICATION AND AUTHORIZATION
Groups
Add
Users
Add
Q1
Categorys
Add
Pages
Add
Q2
Livess
Add
Workss
Add
Select lives to change
Action:

Go
0 of 1 selected
LIVES
Mehul - Noida
1 lives

Q2 - WORKS and LIVES

127.0.0.1:8000/q2/

Q2: WORKS and LIVES

Insert into LIVES

Person name:

Street:

City:

Insert

Insert into WORKS

Person name:

Company name:

Salary:

Insert

Find Employees by Company

Company name: STM

Search

Results

- Mehul - Noida